Bioinformatics Bootcamp 2026

Assessment 01

Covered Lectures:

• Lecture 01: Introduction to Bioinformatics

• Lecture 02: Biological Databases

Instructions:

- Answer all questions in your own words.
- Support your answers with relevant examples.
- Avoid copying from AI tools or online sources.
- Recommended length: 150–300 words per question.

Core Base :

- Bioinformatics is often described as an interdisciplinary field.
- Explain why Bioinformatics requires knowledge from multiple disciplines. Discuss the role of biology, computer science, mathematics, and statistics in solving biological problems.

Question 1

A researcher has discovered a DNA sequence that may be associated with a rare genetic disorder. Before conducting laboratory experiments, they want to determine whether this sequence has been reported previously.

Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides and how it would assist the researcher.

Question 2

A scientist is studying a disease-causing bacterium and wants to compare its genome with those of related bacterial species to identify conserved genes.

Which biological database(s) would you recommend for this analysis? Explain why these databases are appropriate and what information they provide.

Question 3

A biotechnology company has isolated a protein from a newly identified microorganism and wants to investigate whether similar proteins exist in other species.

Which biological database(s) would you recommend, and what information would you expect to obtain from each database? Explain your reasoning.

Question 4

A scientist has identified an unknown protein sequence and wants to determine its function.

Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides.

Question 5

You have been provided with an unknown biological sequence but are not told whether it is DNA, RNA, or protein.

Describe the step-by-step approach you would follow to identify the sequence using biological databases. Explain which databases you would use at each stage and why.

Note:

· **Submission Deadline: Wednesday, 8 July 2026**

· A brief note stating that **outstanding submissions will be nominated for publication on the official BioCode Innovators article page.**

Submission Format: Submit your assessment as a **PDF file**. Your work should reflect **critical thinking**, conceptual understanding, and your own analysis of the topics covered. Well-reasoned and original responses are highly encouraged.